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Abstract
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Toward Improved Soil Moisture Retrieval Using NISAR InSAR Closure Phase Measurements

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Challenge

Surface soil moisture can be retrieved from SAR backscatter data; however, these techniques are sensitive to surface roughness changes and extensive spatial averaging is required. In the last years, a significant amount of work has been done to establish a framework for the retrieval of surface soil moisture from InSAR products, such as the interferometric phase and coherence. It is hereby proposed not to use the interferometric phase directly, since interferometric phase measurement are also sensitive to many other contributors (e.g. atmosphere, deformation, orbital errors), but to use the so-called closure phase. The closure phase is the cyclic product of spatially averaged interferograms between combinations of three SAR acquisitions. The idea is to increase the accuracy of surface soil moisture measurements by using the closure phase method or by combining the closure phase method with amplitude-based techniques.

This study aims to analyse L-band closure phases signatures from NISAR (NASA-ISRO Synthetic Aperture Radar) across various land conditions, including bare land, croplands and forests. The spatiotemporal statistics of L-band closure phase are investigated and comparisons with Sentinel-1 C-band phase closures will be performed to explore the hidden geophysical signals in InSAR closure phases. Previous studies [1], mainly based on C-band observations, have demonstrated the strong potential of using closure phases to retrieve geophysical parameters such as soil moisture and vegetation water content. A better understanding of closure phase signatures across different radar frequencies will improve the interpretation of interferometric phase signals and contribute valuable insights toward the development of novel techniques of soil moisture retrieval.

Methodology

The following methodological pipeline will be followed:

1. *Generate co-registrated SLC stacks. For Sentinel-1 SLC stack, ISCE2 TopStack functionality* [2] *will be used, while for NISAR SLC stack we are going to use ISCE3* [3].
2. *Get meteorological ERA5-Land dataset (precipitation, snow and temperature) for our area of interest and our specific time span.*
3. *The soil moisture estimation pipeline (Figure 1) already introduced in our paper* [1] *and implemented in our open-source software tool InSAR4SM* [4] *will be applied.*
4. *Perform a comparison analysis of closure phase signals and soil moisture estimates from Sentinel-1 and NISAR data vs other data sources (e.g. ground soil moisture observations from ISMN* [5] *at multiple depths).*

Study area and data

Based on a recent email (8th Jan 2026) that we received from the NISAR team, the study area and data that will be used in this study will be defined after the release of the NISAR data (end of February 2026). An Arctic region will be selected for the study needs. We attach a part of the NISAR`s team email:

"NISAR began collecting data according to its science plan in mid-October 2025, acquiring data on the ascending and descending passes of all land and ice-covered surfaces of Earth. The team has been busy calibrating the data and understanding all the characteristics of its many modes and data products. The first set of data products covering broad discipline areas will be released in February 2026, with more to follow. "

Preliminary results

The tool InSAR4SM when applied on arid regions produced soil moisture estimates with satisfactory accuracy (RMSE < 0.04 m^3 / m^3). In this study the methodology will be tested in Arctic, non-frozen, low biomass areas. This section will be completed when NISAR data will be available.

Discussion and Conclusions

Will be provided when NISAR data will be available.

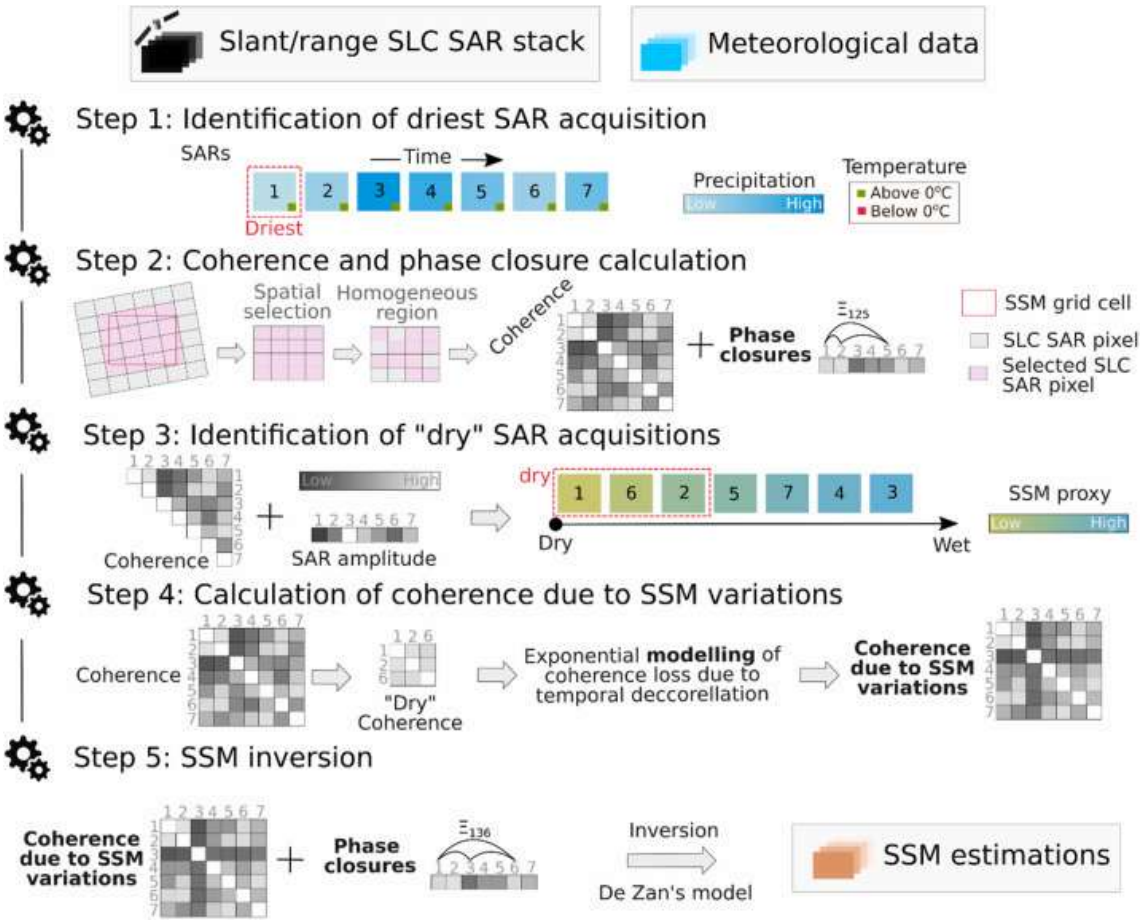


Figure 1 Graphical description of the 5-step proposed methodology. The inputs of the methodology are the slant range SLC stack and meteorological data. The main output of the methodology is the SSM estimations for each image in the SLC stack. Source: [1]

References

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